



Cambridge O Level

CANDIDATE
NAME

--



CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

ADDITIONAL MATHEMATICS

4037/12

Paper 1 Non-calculator

May/June 2026

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

List of formulas

Equation of a circle with centre (a, b) and radius r .

$$(x - a)^2 + (y - b)^2 = r^2$$

Curved surface area, A , of cone of radius r , sloping edge l .

$$A = \pi r l$$

Surface area, A , of sphere of radius r .

$$A = 4\pi r^2$$

Volume, V , of pyramid or cone, base area A , height h .

$$V = \frac{1}{3} Ah$$

Volume, V , of sphere of radius r .

$$V = \frac{4}{3} \pi r^3$$

Quadratic equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

Arithmetic series

$$u_n = a + (n - 1)d$$

$$S_n = \frac{1}{2} n(a + l) = \frac{1}{2} n\{2a + (n - 1)d\}$$

Geometric series

$$u_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1)$$

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

Formulas for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

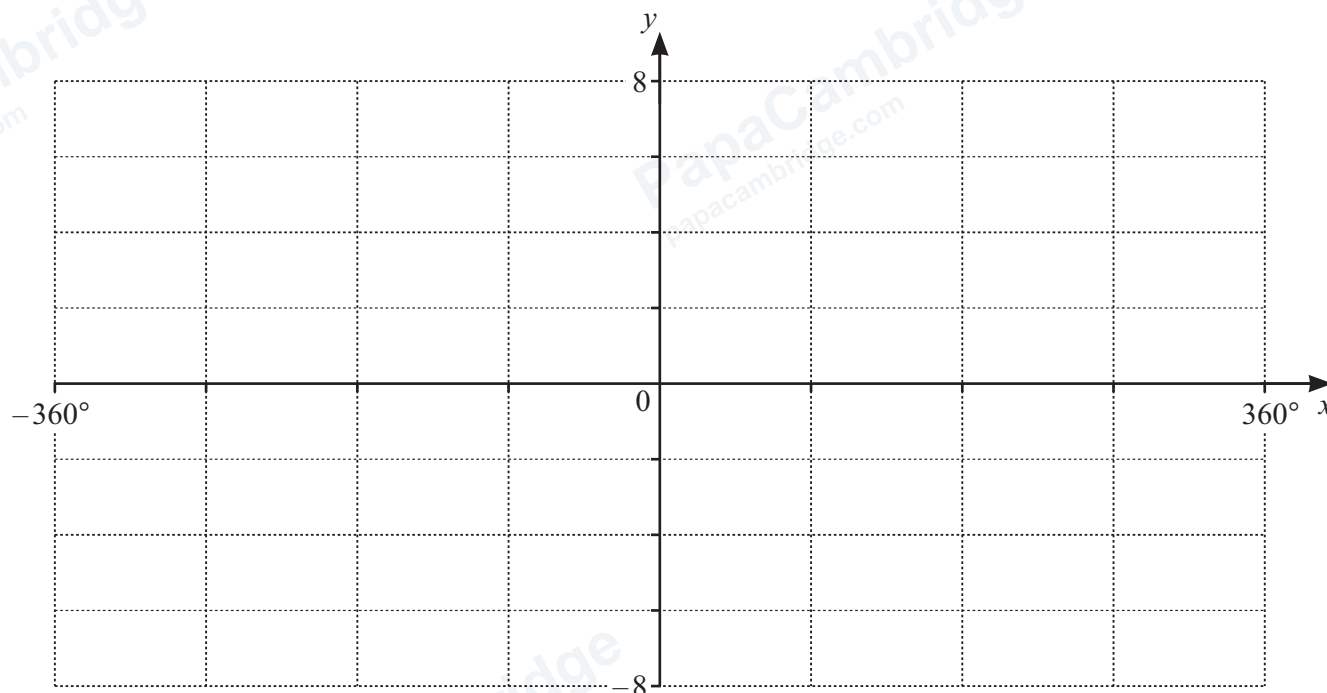
$$\Delta = \frac{1}{2} ab \sin C$$



Calculators must **not** be used in this paper.

- 1 (a) On the axes, sketch the graph of $y = 4 \cos x - 2$ for $-360^\circ \leq x \leq 360^\circ$.

[2]



- (b) Write down the amplitude of $4 \cos x - 2$.

[1]

- (c) Write down the period of $4 \cos x - 2$.

[1]

- (d) Write down the set of values of the constant k for which the equation $4 \cos x - 2 = k$ has at least two solutions.

[1]

- 2 Points A and B have coordinates $A(-2, 3)$ and $B(8, -3)$.

- (a) The perpendicular bisector of AB passes through the point $C(0, c)$.

Find the value of c .

[3]

- (b) The point D has coordinates $(2, -1)$.

The lengths of AD and BD are such that $BD = \frac{\sqrt{n}}{2}AD$ where n is an integer.

Find the value of n .

[4]



3 The polynomial p is given by $p(x) = 6x^3 - x^2 - 5x + 2$.

(a) Find the remainder when $p(x)$ is divided by $x + 2$.

[1]

(b) Write $p(x)$ as a product of linear factors.

[4]

(c) Hence solve the equation $6u^6 - u^4 - 5u^2 + 2 = 0$.

[2]

4 An arithmetic progression has first term $u_1 = 400$ and common difference $d = -2$.

(a) Find the 81st term and the 100th term.

[2]

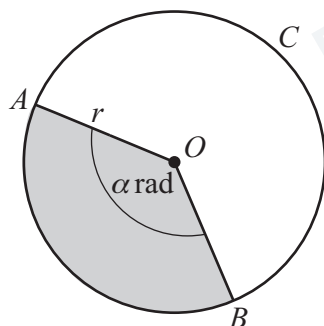
(b) Find the sum of all the terms from the 81st to the 100th, $u_{81} + u_{82} + \dots + u_{100}$.

[3]

- 5 Find, in exact form, the equation of the tangent to the curve $y = \ln((2x-4)^5)$ at the point where $x = 3$.

[4]

- 6 In this question lengths are in centimetres.



The diagram shows a circle with centre O and radius r .
Points A , B and C are on the circumference of the circle.
Angle $AOB = \alpha$ radians.

- (a) The perimeter of the shaded sector is equal to the length of the major arc, ACB .

Find the exact value of α .

[4]

- (b) The area of the shaded sector is $18(\pi^2 - 1)$.

Find the exact value of r .
Simplify your answer.

[4]





7 Show that $\operatorname{cosec} x - \sin x$ can be written as $\frac{\cos x}{\tan x}$.

[3]

8 Variables x and y are related by the equation $y = \frac{5x+1}{x-2}$.

Using calculus, find the approximate change in x when y increases from 4 by the small amount h .

[5]

- 9 (a) Write $\frac{1}{\log_x e} - \ln(x-1)$ as a single logarithm to base e.

[2]

- (b) Solve the equation $2 \log_5 2x = 1 + \log_5 (9x - 10)$.

[5]



10 It is given that a geometric progression has the following terms.

$$\begin{array}{ll} \text{2nd term} & 11x - 1 \end{array}$$

$$\begin{array}{ll} \text{3rd term} & x + 5 \end{array}$$

$$\begin{array}{ll} \text{4th term} & x - 1 \end{array}$$

(a) Given that $x > 0$, find the common ratio and first term of the progression.

[6]

(b) The sum to infinity of this progression is $\frac{k}{3}$ where k is an integer.

Find the value of k .

[2]



- 11 The position vectors of points A and B relative to an origin O are

$$\overrightarrow{OA} = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}.$$

The point C lies on AB and is such that $AC : CB = 3 : 1$.

The point D lies on OA such that $\overrightarrow{OD} = \lambda \overrightarrow{OA}$ and $\overrightarrow{CD} = \begin{pmatrix} 5 \\ -5.5 \end{pmatrix}$.

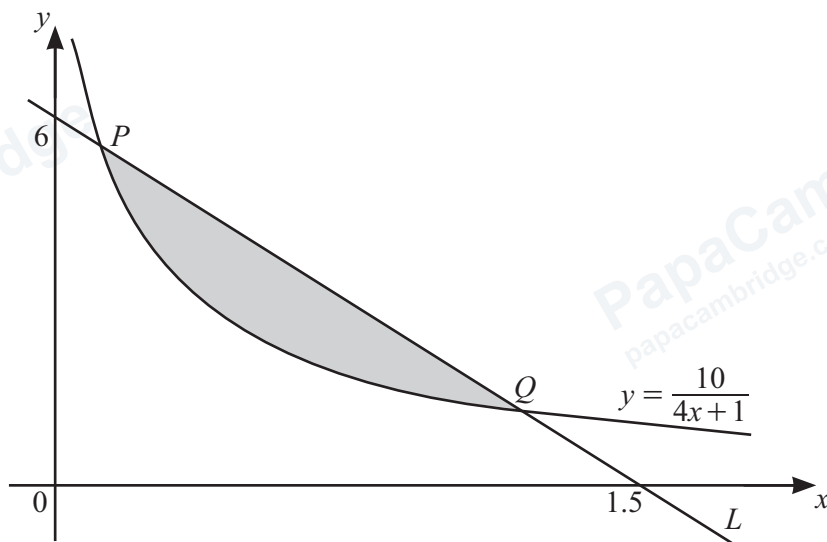
- (a) Find $\overrightarrow{OD}$.

[5]

- (b) Hence find the value of λ .

[1]

12



The diagram shows part of the curve $y = \frac{10}{4x+1}$ and the line L .

L passes through the points $(0, 6)$ and $(1.5, 0)$.
The curve and L intersect at the points P and Q .

Find the exact area of the shaded region.

[11]



Continuation of working space for Question 12.

Question 13 is on the back page.





13 Show that, for all values of $n \geq 3$, ${}^{n+2}C_3 - {}^nC_3 = n^2$.

[4]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

